PART 2: Similarity based on paths

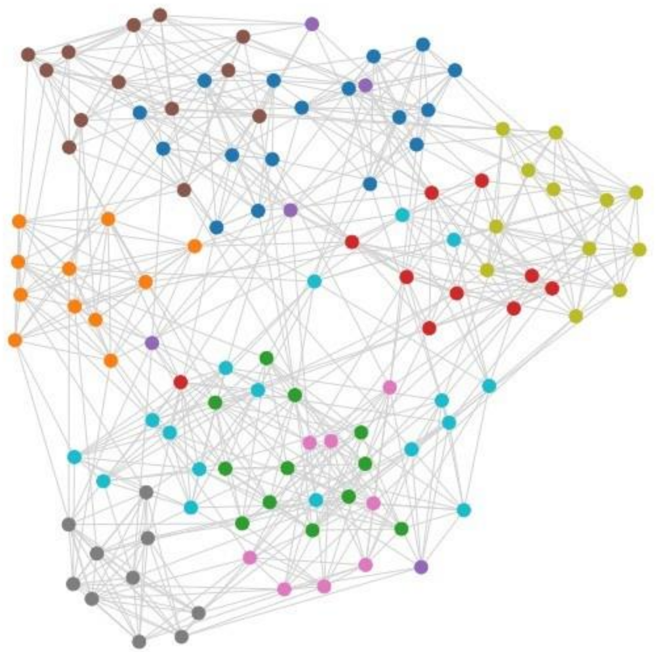
Assumption: Nodes connected by many paths of short distance are more likely to be connected

A: Presence of path between node *i* and *j*
A^2: Number of paths between node *i* and *j* in two steps
A^3: Number of paths between node *i* and *j* in three steps
…

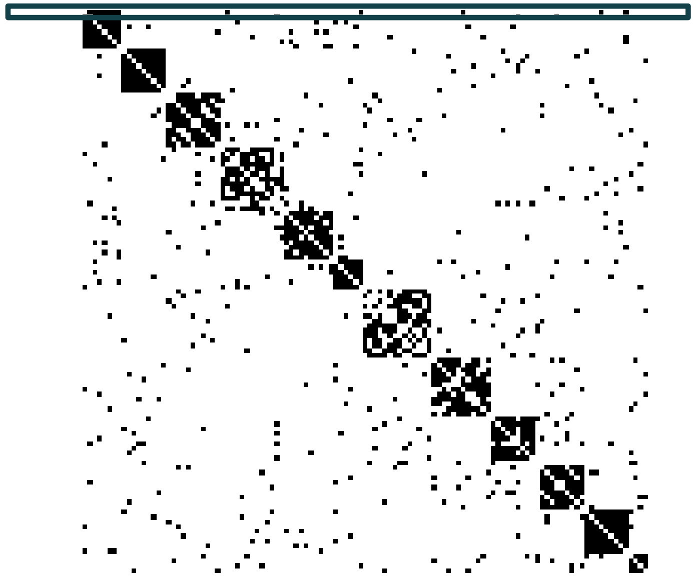
Katz similarity: counts the number of walks at all distances between two nodes, giving shorter walks more weight

$$Katz = \beta A + \beta^2 A^2 + \ldots = \sum_{l=1}^{\infty} \beta^l A^l$$

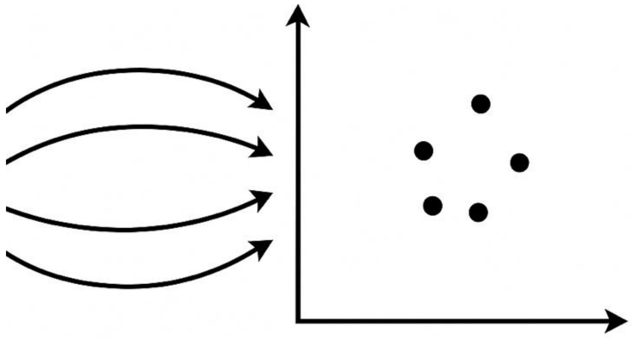
PART 3: Node embeddings



Adjacency matrix



Node embeddings



Can we describe each node with just a handful of numbers?

We can define each node by its connections

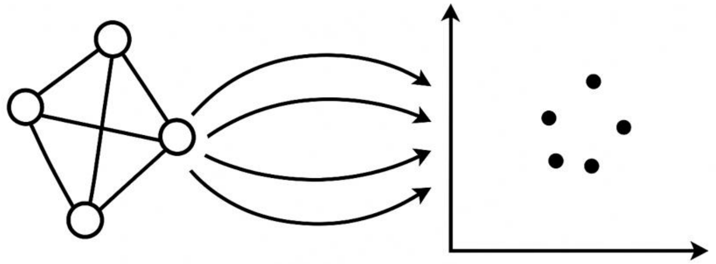
You can use it to predict something about the node:

- but that would mean thousands or millions of parameters!
- and it only provides information at the local level

Lower dimensionality representation

- That captures the network structure
- Where similar nodes should have similar embeddings

Node embeddings



What do we mean by similar nodes?

- Nodes with similar position in the network

  → **Unsupervised learning:** Spectral methods and Shallow neural networks

- Nodes with the same outcome (e.g. criminal nodes in a financial network)

  → **Node classification**

  X = Embedding

  Y = outcome to be predicted

- **Nodes that are connected (for link prediction)**

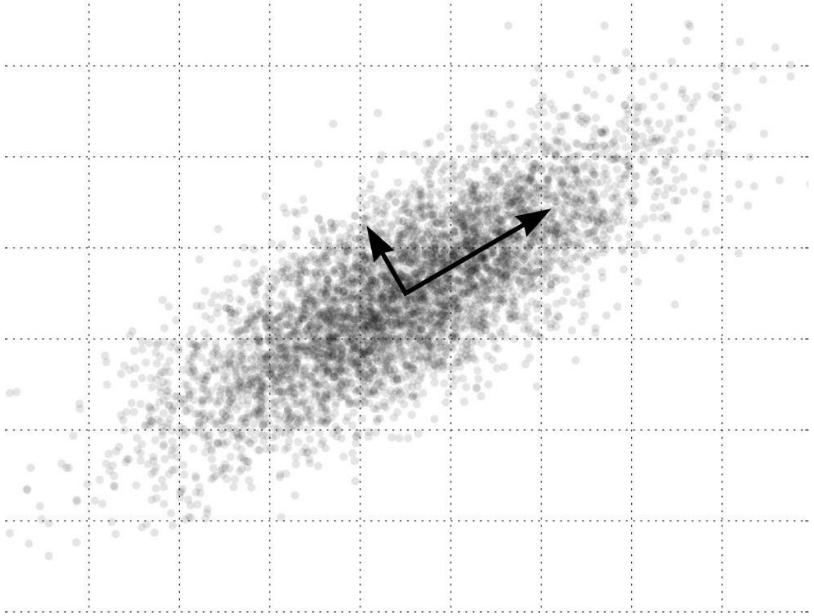
  → **Link prediction**

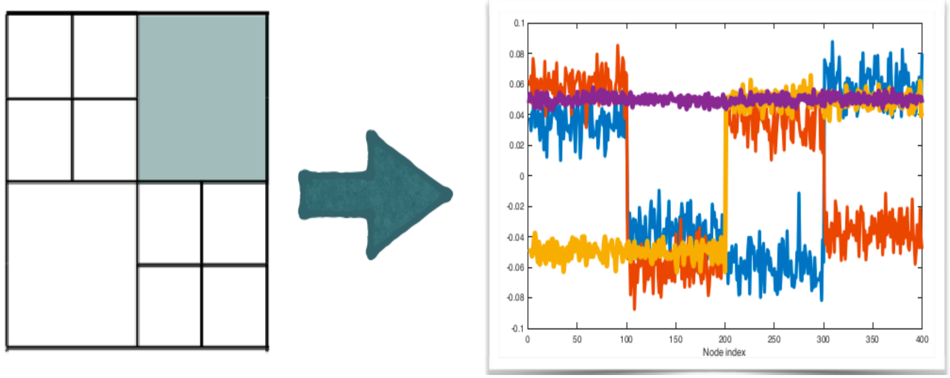
  X = combination of the two embeddings X_1 and X_2 (e.g. X_1*X_2, or np.abs(X_1 - X_2))

  Y = link/no link

Part 3.1: Spectral methods

Related to characteristic eigenvectors of matrices associated with the network





Principal Component Analysis (PCA)

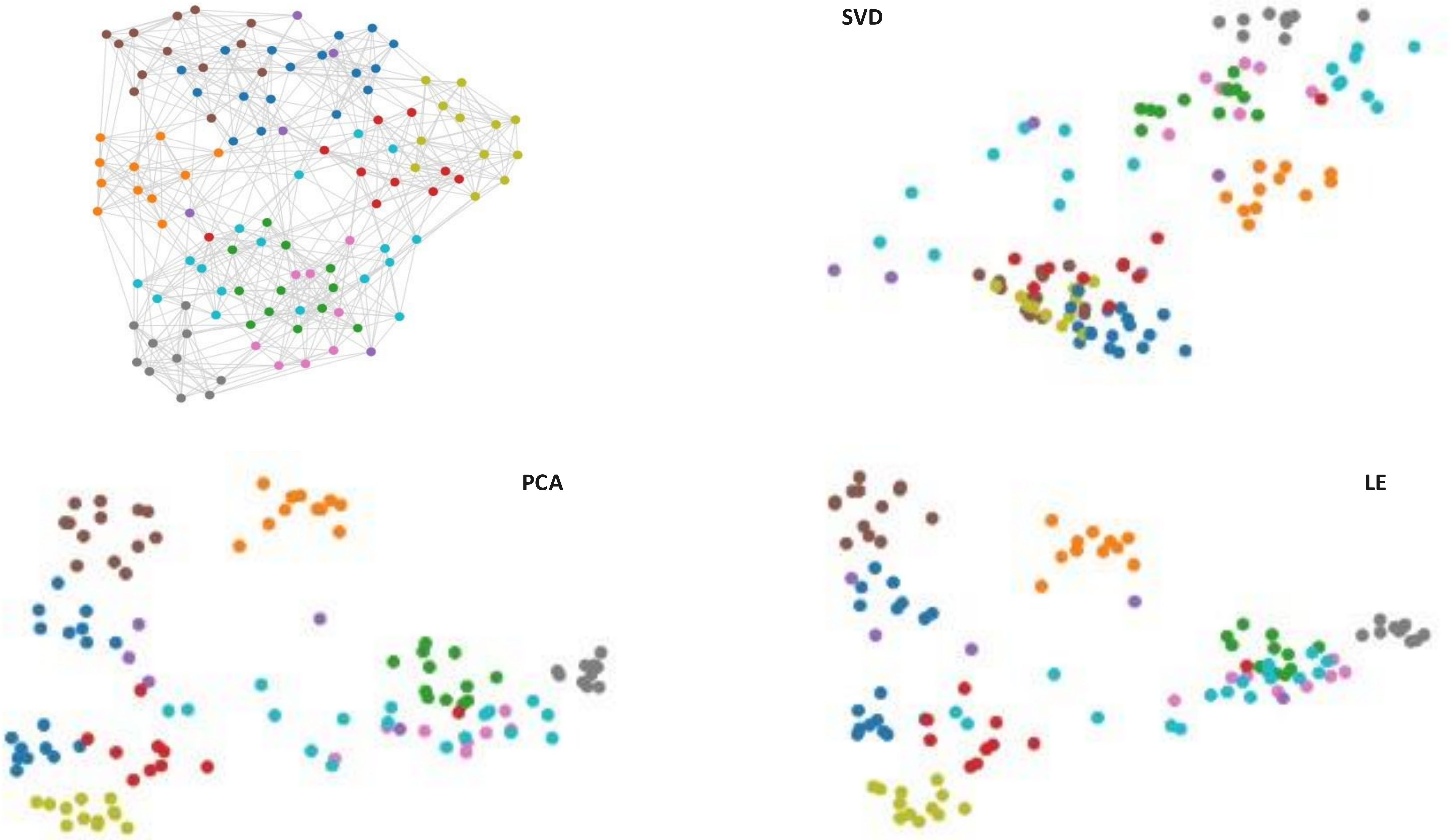
- Correlated variables → Linear combination of orthogonal variables
- Eigenvectors corresponding to the largest k eigenvalues of A^TA (centered)

Singular Value Decomposition (SVD)

- Eigenvectors corresponding to the largest k eigenvalues of A^TA (uncentered)

Laplacian Eigenmaps (LE)

- Assumes that the nodes lie on a low-dimensional space
- Nodes close in that space are also close in the network
- That space = the eigenvectors corresponding to the smallest k eigenvalues of the normalized Laplacian matrix ($I – D^{-1/2}A\, D^{1/2}$)

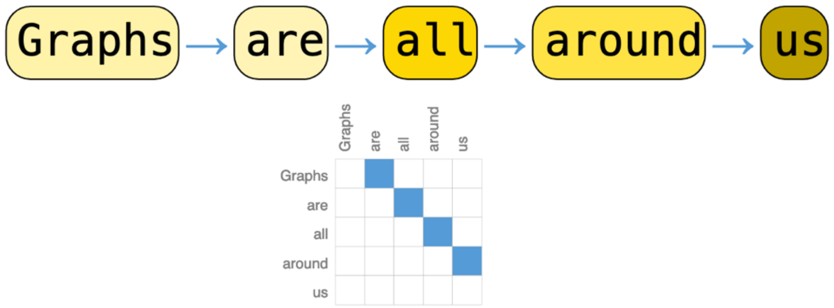


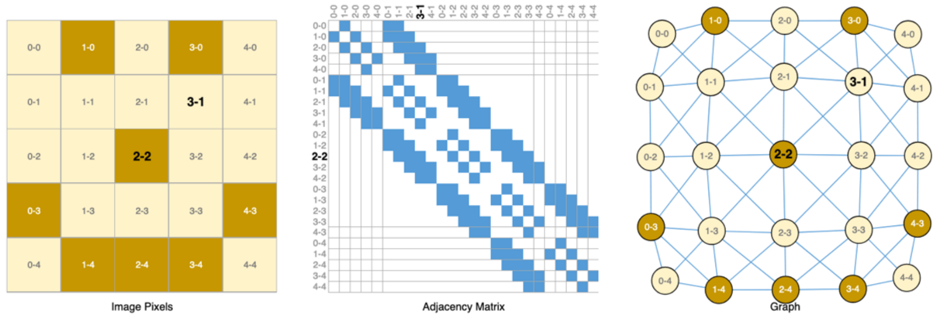
https://javier.science/marimo_node_embeddings/

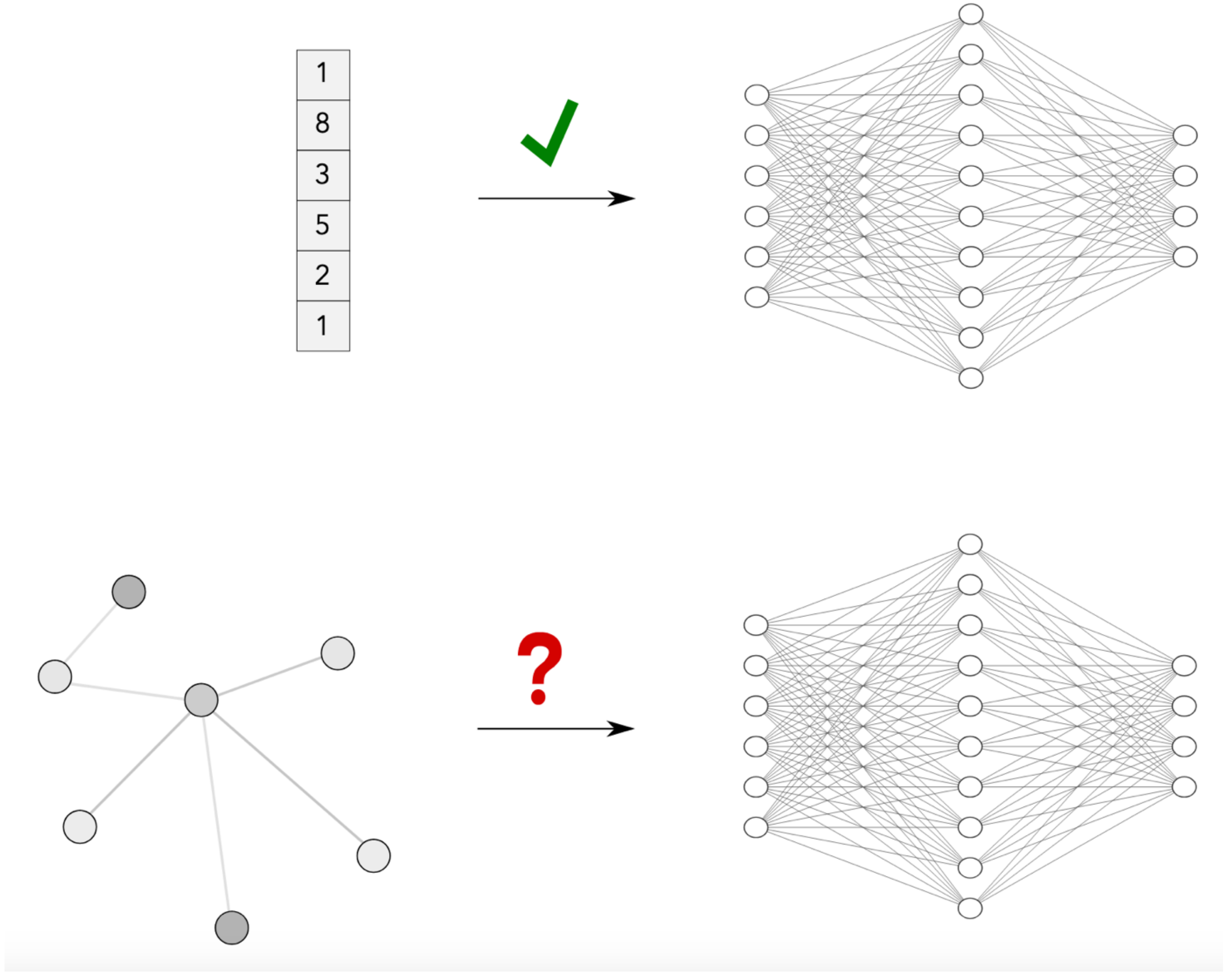
Part 3.2: Shallow Neural Networks

Regular networks: Text, images

- Text analysis: Chain (nodes = words)
- Images: Lattices (nodes = pixels)





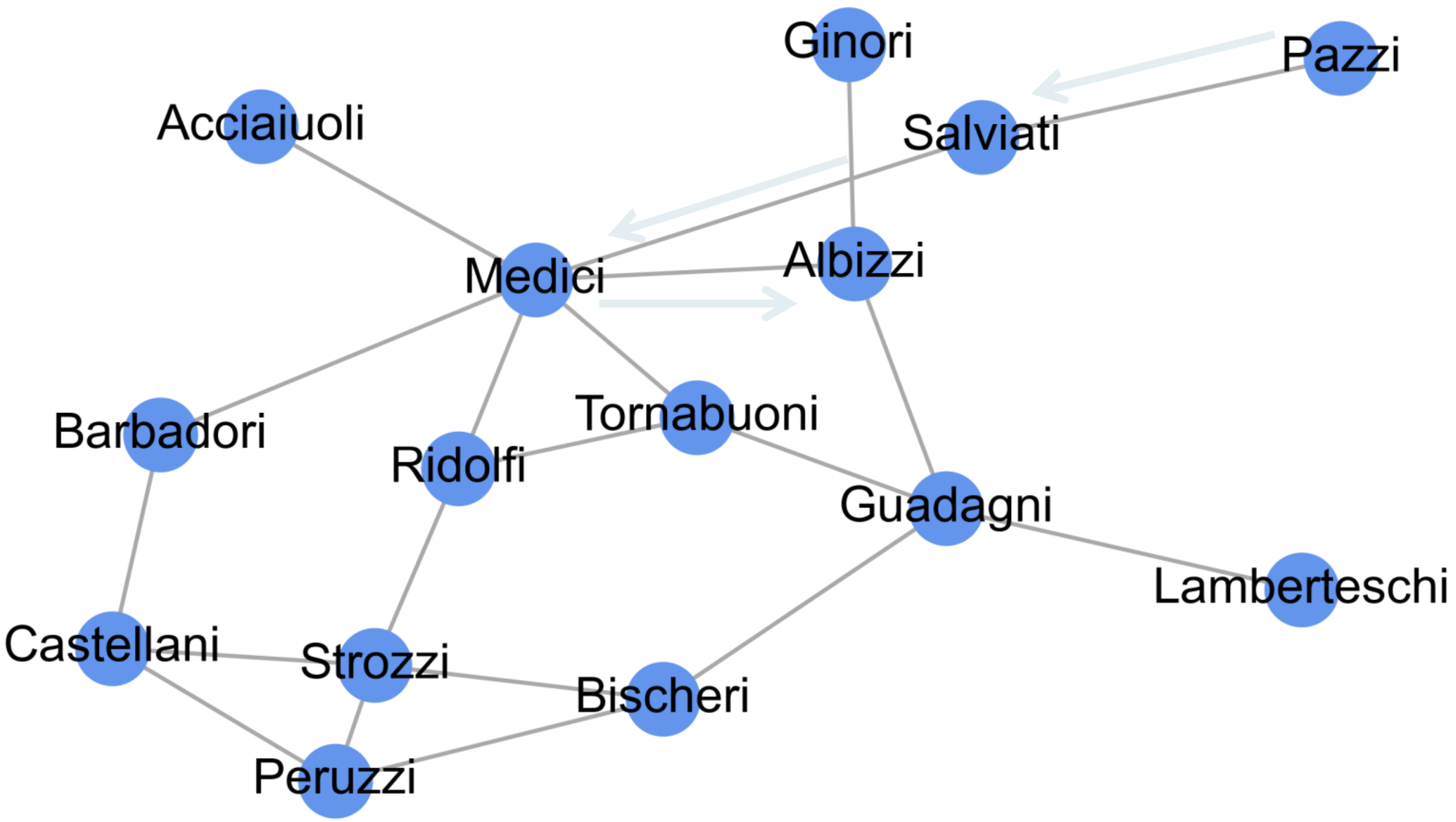


https://distill.pub/2021/understanding-gnns/

Node2vec / deepwalk

Idea:

1. Generate “sentences” using random walks.
2. Use methods from text analysis



Pazzi -> Salviati -> Medici -> Albizzi

…

Text analysis: Word2vec

Distributional hypothesis: similar words will be surrounded by similar words (you will know a word by the company it keeps)

What words tend to appear around “Network”?

- Network Science
- Network Analysis

→ Words *science* and *analysis* are similar

Text analysis: Word2vec

Step 1: Create co-occurrence matrix

- I like deep learning
- I like NLP
- I enjoy flying

Context word

Target word

counts	I	like	enjoy	deep	learning	NLP	flying	.
I	0	2	1	0	0	0	0	0
like	2	0	0	1	0	1	0	0
enjoy	1	0	0	0	0	0	1	0
deep	0	1	0	0	1	0	0	0
learning	0	0	0	1	0	0	0	1
NLP	0	1	0	0	0	0	0	1
flying	0	0	1	0	0	0	0	1
.	0	0	0	0	1	1	1	0

[Word Sense Disambiguation Using Vector Space Models]

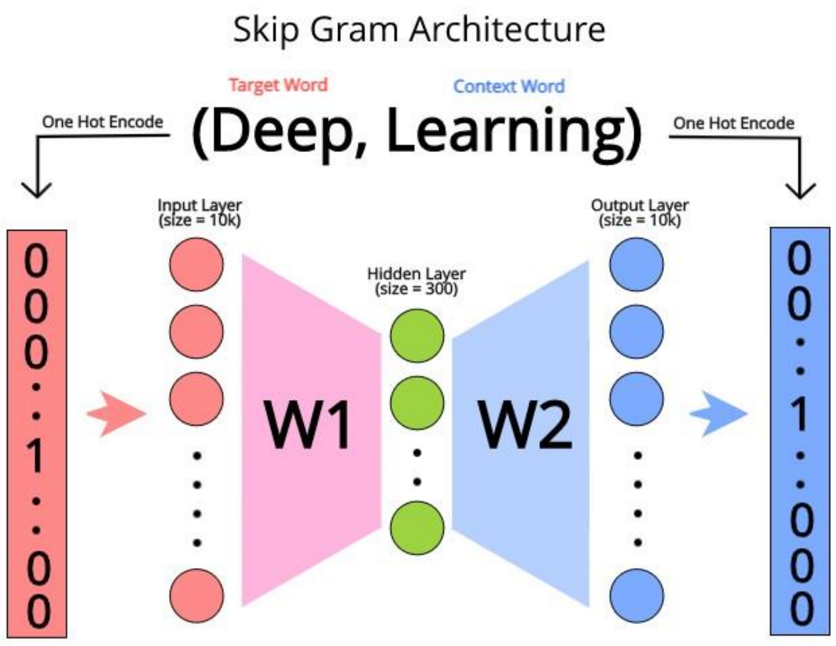
Text analysis: Word2vec

Step 2: Train a classification model

- *Positive examples*: target word should predict context words (Skip-Gram)
- *Negative examples*: target word should not predict random words (i.e., not context words)

Two vectors are similar if they have a high dot product (~cosine similarity)

- Vector associated to “deep”: w1[“deep”,:]
- Vector associated to “learning”: w2[:,”learning”]



Intuition:

- Modify W1 and W2 so target embeddings are close (have a high dot product) to context embeddings for nearby words and further from context embeddings for noise words that don’t occur nearby

Speech and language processing: an introduction to natural language processing, computational linguistics, and speech recognition (Jurafsky and Dan, 2009)

In networks: DeepWalk / Node2Vec

Distributional hypothesis: The more often two nodes appear near the same nodes in the same random walk, the more similar their embeddings will be.

Step 1: Create co-occurrence matrix

Pazzi -> Salviati -> Medici -> Albizzi

Medici -> Albizzi -> Guadagni -> Medici

Context node

Target node

…	Pazzi	Salviati	Guadagni	Medici	Albizzi
Pazzi		1		1	1
Salviati	1				
Guadagni					
Medici	1		1	1	1
Albizzi	1			1	

Step 2: Create embeddings representing how similar the neighbors of each node are

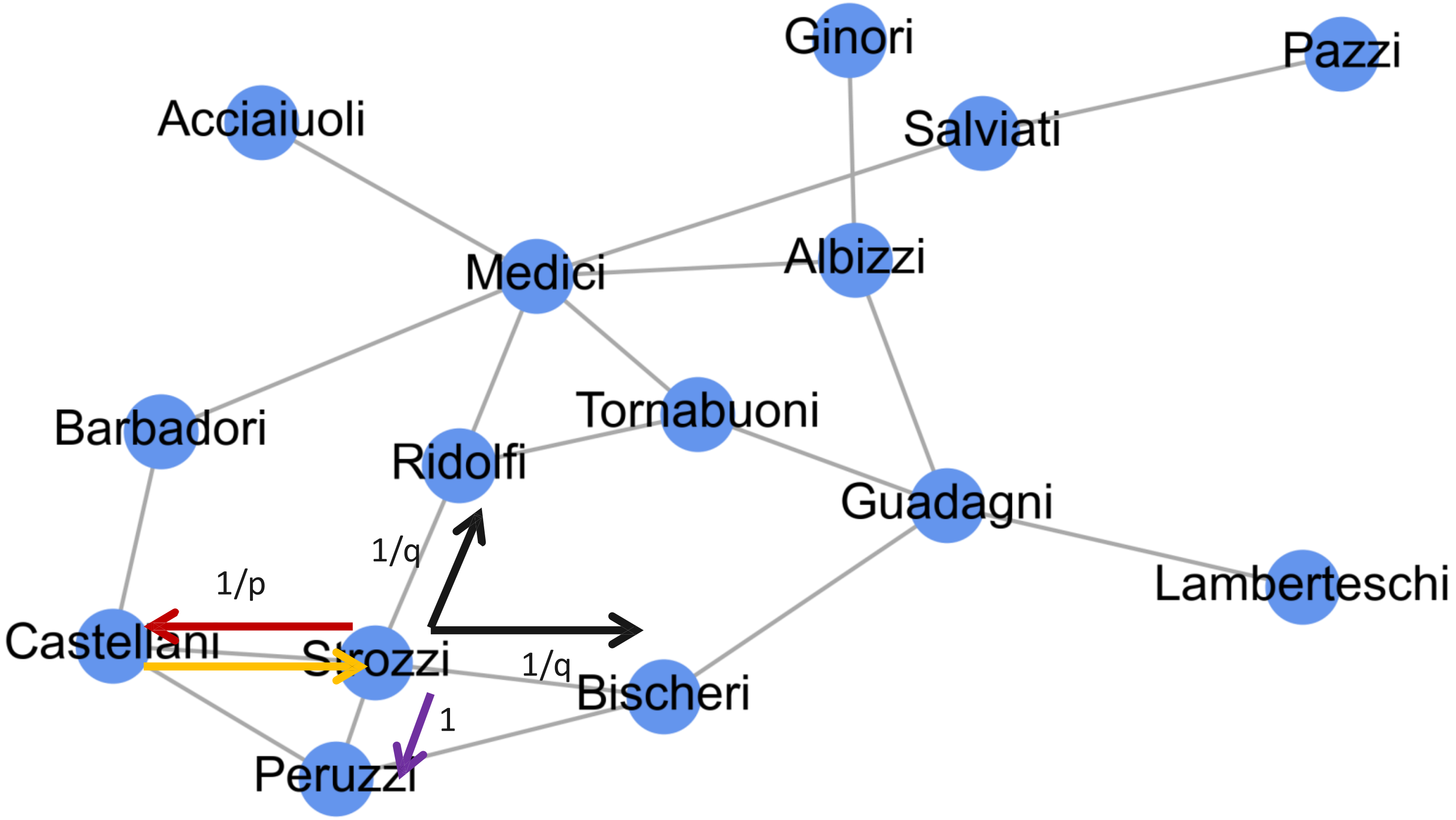
node2vec

Difference with deepwalk: generate “sentences” using biased random walks.

p = controls probability of going back to previous node

1 = weight of going to a node adjacent to the previous one

q = controls probability of going to new nodes



Using node2vec

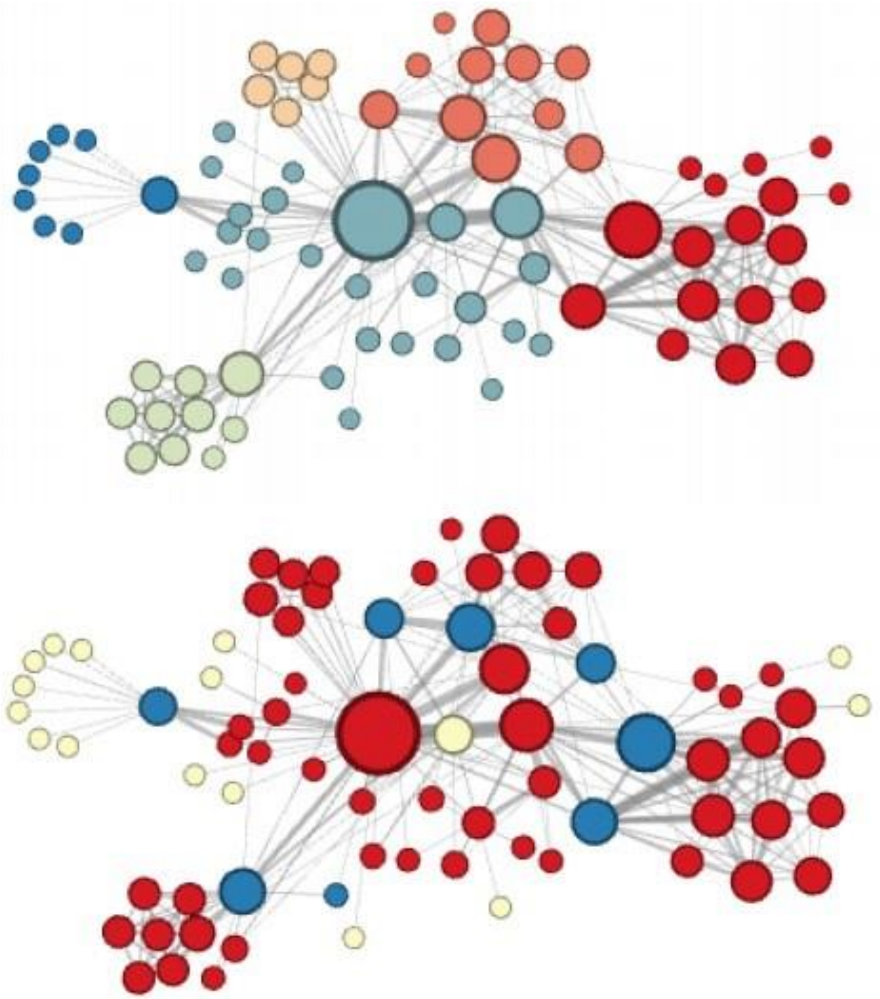


Figure 3: Complementary visualizations of Les Misérables coappearance network generated by *node2vec* with label colors reflecting homophily (top) and structural equivalence (bottom).

Depending on q

~ similarity reflecting clusters

~ similarity reflecting "structural roles"

Big problem: how to set up q and p?

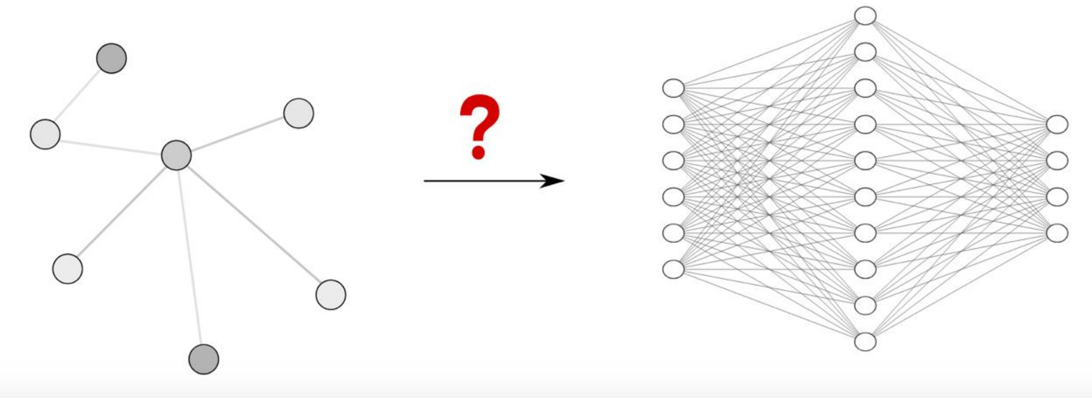
→ Or learn embeddings for the task directly → GNNs

Part 3.3: Graph Neural Networks (GNNs)

Node2vec created the embeddings in an unsupervised (or self-supervised) way. But we can do it in a *supervised* way, so **node embeddings are similar if nodes have the same outcome**

In the context of link prediction → The same outcome = being connected

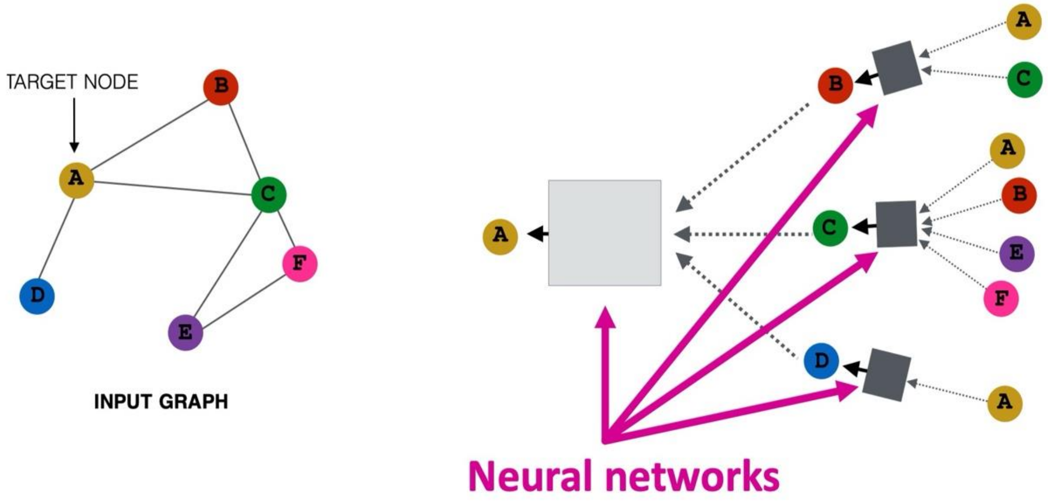
Problem:



https://distill.pub/2021/understanding-gnns/

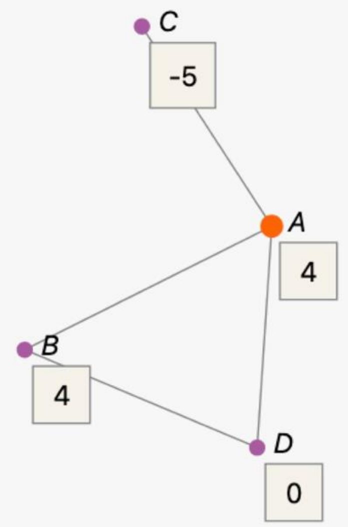
Solution:

- **Intuition:** Nodes aggregate information from their neighbors using neural networks

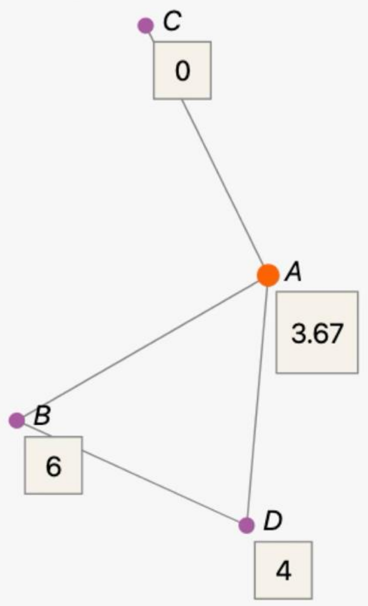


https://web.stanford.edu/class/cs224w/

GNN: Graph Neural Networks ~ message passing + feature mixing



$$
\begin{aligned}
h_A^{(1)} &= f\left(W^{(1)} \times \frac{h_B^{(0)} + h_C^{(0)} + h_D^{(0)}}{3} + B^{(1)} \times h_A^{(0)}\right) \\
&= f\left(1 \times \frac{4 + -5 + 0}{3} + 1 \times 4\right) \\
&= f\left(-0.33 + 4\right) \\
&= f\left(3.67\right) \\
&= \mathrm{ReLU}\left(3.67\right) = 3.67.
\end{aligned}
$$



$$
\begin{aligned}
h_A^{(2)} &= f\left(W^{(2)} \times \frac{h_B^{(1)} + h_C^{(1)} + h_D^{(1)}}{3} + B^{(2)} \times h_A^{(1)}\right) \\
&= f\left(1 \times \frac{6 + 0 + 4}{3} + 1 \times 3.67\right) \\
&= f\left(3.33 + 3.67\right) \\
&= f\left(7\right) \\
&= \mathrm{ReLU}\left(7\right) = 7.
\end{aligned}
$$

$$\mathbf{H}^{(\ell+1)} = \sigma\left(\hat{\mathbf{A}}\mathbf{H}^{(\ell)}\mathbf{W}^{(\ell)}\right)$$

Final embedding used to predict college completion

https://distill.pub/2021/understanding-gnns/

Recap node embeddings

We want to create low-dimensional embeddings:

- Spectral methods → Based on the eigenvectors of the network
  - Unsupervised, similarity = similar position in the network

- Shallow neural networks → Based on random walks on the network
  - Unsupervised/self-supervised, similarity = similar neighborhood

- Graph Neural Networks → Based on message passing between nodes + node features
  - Supervised, similarity = similar position in the network and similar outcome
  - Many GNNs (GCN, GAT, SAGE, Transformer…)

Now: Predict links in the PPI network